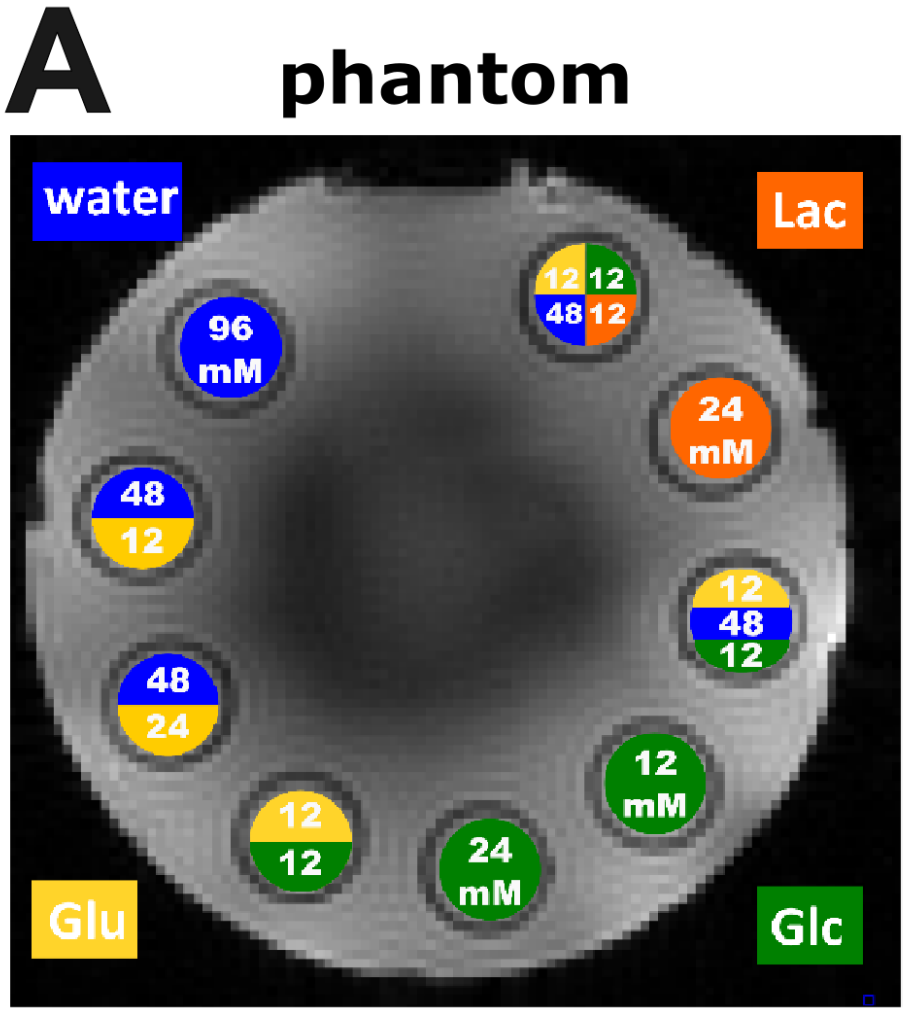
Spherical Deuterium metabolite phantom

Spherical phantom (18 cm diameter) filled with hydrogel of PBS and tubes 1- 9 with 2H metabolites arranged as shown below.



Stock solutions:

- **Stock0** - 300 mL PBS + 300 mg NaN3
- **Stock1** - 100 mL of 24 mM Glu
  - 100mL Stock0 + 445.44 mg DL-Glutamic acid-d2
  - Acidic solution is not good for gelation : increase pH to ~7.2 with 8M and 1M NaOH
- **Stock2**- 100 mL of 24 mM of Glc
  - 100mL Stock0+ 437.21 mg D-Glucose-6,6-d_2
- **Stock3** - 50 mL of 24 mM Lac
  - 50mL Stock0 + 307 µL Na-lactate-3,3,3-d3 (solution, 45-55% (w/w) in water)

Tube No	Labels	content	preparation	Remarks
1	D20	PBS + 96 mM D20 + 31 g/L cellulose	30 mL Stock0 + 47.26 uL (52.46 mg) D20 + 930 mg hydroxyethylcellulose	D20 is of 4x other metabolite added =96-8.667= 87.33 mM D20 deuterium density: 1.11 g/cm^3 at 20 °C
2	D20+Glu	PBS + 48 mM D20 +12 mM Glu +0.1 % NaN3 + cellulose	15 mL Stock0 + 15 mL Stock1 + 21.28 uL (23.63 mg) D20 + 930 mg hydroxyethylcellulose	added =48-8.667= 39.33 mM D20
3	D20+ Glu	PBS + 48 mM D20 +24 mM Glu +0.1 % NaN3 + cellulose	30mL Stock1 + 21.28 uL (23.63 mg) D20 + 930 mg hydroxyethylcellulose	added =48-8.667= 39.33 mM D20 In the original recipe, 48 mM D20 was not planned.
4	Glu+Glc	PBS +12 mM Glu + 12 mM Glc +0.1 % NaN3 + cellulose	15 mL Stock1 + 15 mL Stock2 + 930 mg hydroxyethylcellulose	
5	Glc	PBS +24 mM Glc +0.1 % NaN3 + cellulose	30 mL Stock2 + 930 mg hydroxyethylcellulose	
6	Glc	PBS +12 mM Glc +0.1 % NaN3 + cellulose	15 mL Stock0 + 15 mL Stock2 + 930 mg hydroxyethylcellulose	In the original recipe, 48 mM D20 was planned but probably added to vial 3.
7	D20+Glc +Glu	PBS +12 mM Glc +12 mM Glu + 48 mM D20 +0.1 % NaN3 + cellulose	15 mL Stock1 + 15 mL Stock2 + 21.28 uL (23.63 mg) D20 + 930 mg hydroxyethylcellulose	added =48-8.667= 39.33 mM D20
8	Lac	PBS +24 mM lac +0.1 % NaN3 + cellulose	30 mL Stock3 + 930 mg hydroxyethylcellulose	we need ~200 mg of Lac

9	Lac+Glu	PBS +12 mM Lac +12 mM Glc + 12 mM Glu + 48 mM D2O +0.1% NaN3 + cellulose	15 mL Stock3 15 mL Stock0 + 65,58 mg Glc + 66,82 mg Glu + 21.28 uL (23.63 mg) D20 + 930 mg hydroxyethylcellulose	closer spectral peak because of the glutamic acid the pH of the mixture has to be adjusted to ~7.3 with 8M and 1M NaOH before adding hydroxyethylcellulose
outside	-	PBS + cellulose + 0.1% NaN3	3 L PBS (2x 1.5 L) + 93 g hydroxyethylcellulose + 3 g NaN3	everything has a natural abudance of D20(**0.0156%**) = 8.667 mM

Materials

- ~20 mL Container + 10 mL (gelation wastage)
- transparent sphere (14 cm diameter, 18 cm diameter)
- PBS
- NaN3 (anti-microbial) (0.1 % w/v)
- Hydroxyethylcellulose (31 g/L)

Chemicals	**molar weight [g/mol]**	**price**	**link**
D2O	20.023	~4 $/ml	link
D-Glucose-6,6-d_2	182.17	170$/g	link
DL-Glutamic acid-4,4-d2 + HCl	185.60	>400 $/g	link?
Sodium Lactate	115.08	1.4k/g	
NaN3	65.01	cheap	link
Hydroxyethylcellulose	?	cheap	link